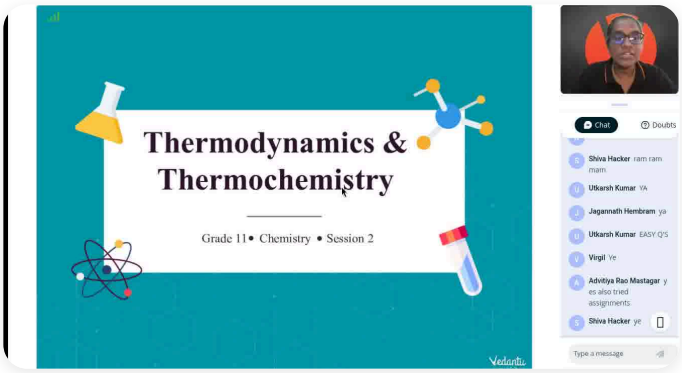
Notes

Introduction to Thermodynamics and Session Overview



Welcome and Recap

We are starting a new chapter on Thermodynamics. In the previous class, we discussed basic terminologies used in thermodynamics. It is important to revise these concepts.

Today's Topics

Today's session will cover the following key concepts:

- First Law of Thermodynamics
- Zeroth Law of Thermodynamics (briefly at the end)
- Internal Energy
- Heat and Work
- Calculation of Work

We will also touch upon reversible and irreversible work, heat capacity, adiabatic processes, and Poisson's ratio, though some of these might be covered in subsequent sessions.

Assignment Details

Assignments will be provided, and it is crucial to complete all questions after each session to prepare for JEE Mains. Today's assignment will be divided into two halves, with specific question numbers to be solved based on the topics covered in this session. The remaining questions will be addressed after covering topics like adiabatic work, Poisson's ratio, and heat capacity.

Recap of Basic Thermodynamic Terminologies

Key Definitions

- **System:** The part of the universe under thermodynamic study.
- **Surrounding:** Everything outside the system that can interact with it.
- **Boundary:** The real or imaginary surface separating the system from the surroundings.
- **Universe:** The combination of the system and its surroundings. Universe = System + Surrounding

Types of Systems

Type of System	Heat Exchange	Mass Exchange
Open System	Yes	Yes
Closed System	Yes	No
Isolated System	No	No

Introduction to Fundamental Thermodynamic Qualities

Fundamental Qualities of Thermodynamics

In thermodynamics, several fundamental qualities are crucial for understanding energy transformations. The primary concepts we will begin with are:

- **Internal Energy**
- **Heat**
- **Work**

These three concepts form the basis for understanding more advanced thermodynamic principles. While other important concepts like enthalpy and Gibbs free energy exist, a strong grasp of internal energy, heat, and work will make understanding those concepts much easier.

Internal Energy (U or E)

Definition

Internal energy is the sum of all energies present within a system. It includes various forms of energy such as kinetic energy (translational, rotational, vibrational), potential energy

Notes

State Function

Internal energy is a **state function**. This means its value depends only on the current state of the system (e.g., temperature, pressure, volume, composition) and not on the path taken to reach that state.

Calculation of Internal Energy

It is difficult to calculate the absolute value of internal energy (U) of a system at a particular instant. However, the **change in internal energy** (ΔU) can be calculated. In thermodynamics, we primarily deal with changes in internal energy.

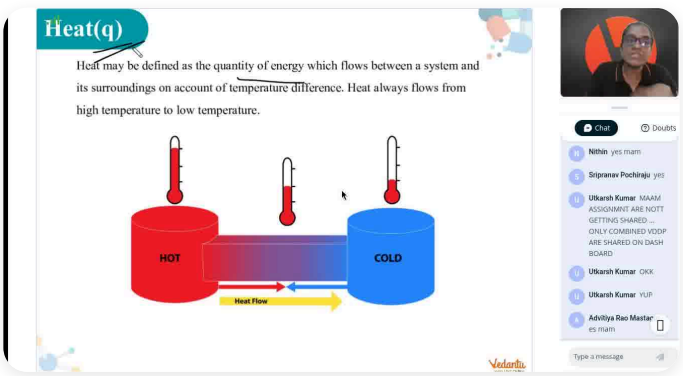
$$\Delta U = U_{\text{final}} - U_{\text{initial}}$$

Where:

- ΔU represents the change in internal energy.
- U_{final} is the internal energy of the system in its final state.
- U_{initial} is the internal energy of the system in its initial state.

Internal energy is sometimes represented by the symbol U and sometimes by E.

Heat (q)



Definition of Heat

Heat may be defined as the quantity of energy which flows between a system and its surroundings on account of a temperature difference. Heat always flows from a region of high temperature to a region of low temperature.

Sign Convention for Heat (in Chemistry)

In thermodynamics, specifically in chemistry, a conventional system is used to represent heat (Q):

- **Endothermic Process:** When heat is absorbed by the system, Q is considered positive ($Q > 0$).

Notes

Work Done in Thermodynamics

In thermodynamics, the third key concept we will explore, after internal energy and heat, is **work done**.

Classical Definition of Work

Traditionally, work done is defined as: Work Done = Force × Displacement

Work Done in Chemical Thermodynamics

In chemical thermodynamics, our focus is primarily on **gases** and **chemical reactions**. The laws and principles we study are mostly applied to these systems.

Chemical thermodynamics can be broadly divided into two parts:

1. **Thermodynamics**: This part largely deals with work done, especially during processes like the expansion or compression of a gas under various conditions (e.g., adiabatic, isothermal).
2. **Thermochemistry**: This part focuses on chemical reactions, equations, and the energy changes associated with them.

Our main objective is to understand how the classical definition of work (force × displacement) can be applied to calculate work done during the compression or expansion of a gas in a thermodynamic system.

Units of Work

Understanding Work and its Units

Work done can be expressed in various units, primarily related to energy or force and displacement.

- **Standard Units:**
  - Joules (J)
  - Newton-meter (N·m) - equivalent to Joules.
- **Units for PV Work (Pressure-Volume Work):** When dealing with pressure-volume work, common units include:
  - Liter-atmosphere (L·atm)
  - Liter-bar (L·bar)
- **Other Units:**

It is crucial to know the conversion factors between these units:

- 1 Joule (J) = 1 Newton-meter (N\cdotpm)
- 1 Liter-atmosphere (L\cdotpatm) = 101.3 Joules (J)
- 1 Liter-bar (L\cdotpbar) = 100 Joules (J)

Sign Convention for Work Done

Work Done by the System vs. Work Done on the System

The sign convention for work is critical in thermodynamics:

- **Work Done by the System:**
  - When the system performs work (e.g., a gas expanding and pushing against its surroundings), it expends energy. This is considered a negative process for the system.
  - Therefore, work done by the system is always **negative** ($W < 0$).
- **Work Done on the System:**
  - When work is done on the system (e.g., a gas being compressed by external pressure), the system gains energy. This is considered a positive process for the system.
  - Therefore, work done on the system is always **positive** ($W > 0$).

Types of Work in Chemistry

Different Forms of Work

While various types of work exist (e.g., gravitational, electrical, mechanical), in the context of chemistry, particularly in thermodynamics, we primarily focus on:

- **Mechanical Work:** This is the most relevant type of work discussed, specifically concerning gases.
  - **Pressure-Volume (PV) Work:** This refers to the work done during the compression or expansion of a gas. It is a fundamental concept in understanding energy changes in chemical reactions and physical processes involving gases.

Types of Work Done Questions

Introduction to Work Done Calculations

In chemistry, particularly in thermodynamics, calculating work done is a crucial concept. Questions related to work done can generally be categorized into three main types:

work?"
    - They then require you to calculate the work done for a specific process, given changes in pressure and volume. For instance, "Calculate the work for this particular process, for pressure change, for this particular volume change."

2. **Graphical Questions:**
    - **Calculating Work from Graphs:** You might be provided with a pressure-volume (P-V) graph and asked to calculate the work done during the process depicted.
    - **Comparing Work Done from Graphs:** Alternatively, you could be given a graph showing multiple processes and asked to identify which process involves more or less work done.
3. **Specific Process Work Done Questions:**
    - These questions focus on calculating work done for particular thermodynamic processes.
    - Examples include: "Calculate the adiabatic work done" or "For an isothermal process, what is the work done?"
    - Further classifications within this type involve distinguishing between reversible and irreversible work.

Classification of Questions

Broadly, the questions you encounter will be of two types:

- **Numerical Type:** Involving direct calculations using formulas.
- **Graph Type:** Requiring interpretation and calculation from P-V diagrams.

Understanding these concepts is fundamental for both numerical and graphical problems related to work done.

Work Done During Gas Compression

Understanding Compression

When a gas is compressed, an external pressure is applied to it. Compression occurs when the external pressure becomes greater than the internal pressure of the gas.

Work Done on the System

During compression, work is done *on the system* (the gas). This is a crucial concept to remember in thermodynamics.

Work, Force, and Displacement

- The **force** applied is the **external pressure** (P_{ext}).
- The **displacement** is the **change in volume** (ΔV) of the gas.

Therefore, for gas compression, the work done can be related to pressure and change in volume.

Work Done in a Pressure-Volume (PV) Graph

Understanding Work Done from a PV Graph

When analyzing a pressure-volume (PV) graph, the work done by or on a system can be determined by the area under the curve. This concept is particularly important when the external pressure is constant.

Constant External Pressure

Consider a process where the system moves from an initial volume (V_i) to a final volume (V_f) under a constant external pressure (P_{ext}).

- **Graphical Representation:**
  - On a PV graph, if the external pressure is constant, the process is represented by a horizontal line at the value of P_{ext}.
  - The initial state is at (V_i, P_{ext}) and the final state is at (V_f, P_{ext}).
  - Connecting these points forms a line segment.
- **Work Calculation:**
  - The formula for work done against a constant external pressure is given by: $W = -P_{ext}\Delta V$ where $\Delta V = V_f - V_i$.
  - Graphically, this corresponds to the area of a rectangle.
    - The height of the rectangle is the constant external pressure (P_{ext}).
    - The width of the rectangle is the change in volume (ΔV).
  - Therefore, the product of P_{ext} and ΔV represents the area below the curve (or line segment) on the PV graph.

General Principle

Always, for any process depicted on a PV graph, the work done is equal to the area below the curve. This principle holds true whether the external pressure is constant or variable. If the pressure is constant, the area is a simple rectangle. If the pressure varies, the area can be calculated using integration or by dividing the area into smaller shapes.

Notes

In thermodynamics, work done can be broadly classified into two types:

- **Reversible Work:** This refers to a process that can be reversed by an infinitesimal change in the conditions. In a reversible process, the system is always in equilibrium with its surroundings. This type of work is theoretical and represents the maximum possible work that can be obtained from a system or the minimum work required to bring about a change.
- **Irreversible Work:** This refers to a process that cannot be reversed by an infinitesimal change in the conditions. Most natural processes are irreversible. In an irreversible process, the system is not in equilibrium with its surroundings, and there is a net loss of energy (usually as heat) to the surroundings. The work done in an irreversible process is always less than the work done in a reversible process for expansion, and more for compression.

Before delving deeper into the specifics of reversible and irreversible work, we will first practice solving some problems based on the concepts covered so far.

Work Done in Gas Expansion

Problem Statement

A gas expands by $0.25\ m^3$ at a constant pressure of $10^3\ N/m^2$. Calculate the work done.

Given Data

- Change in volume (ΔV) = $0.25\ m^3$
- Constant external pressure ($P_{external}$) = $10^3\ N/m^2$

Formula for Work Done

In chemistry, the work done (W) during expansion or compression of a gas at constant external pressure is given by: $W = -P_{external} \times \Delta V$

Sign Convention for Work Done

- **Work done by the system (expansion):** If the system (gas) expands, it does work on the surroundings. By convention, this work is considered **negative**.
- **Work done on the system (compression):** If the surroundings do work on the system (gas) to compress it, this work is considered **positive**.

In this problem, the gas **expands**, which means work is done **by the system**. Therefore, the work done should be negative.

Calculation

Conclusion

The work done by the gas during expansion is -250 J. This corresponds to option B.

Quiz: Work Done During Gas Expansion

Problem Statement

An ideal gas is allowed to expand from 1 liter to 10 liters against a constant external pressure of 1 bar. The work done in joules is:

Key Conversion

Remember that 1 liter bar is equal to 100 joules.

Solution Steps

1. **Formula for Work Done:** The work done (W) against a constant external pressure is given by the formula: $W = -P_{ext} \times \Delta V$ Where:
   - P_{ext} is the constant external pressure.
   - ΔV is the change in volume, calculated as $V_{final} - V_{initial}$.
2. **Identify Given Values:**
   - Initial volume ($V_{initial}$) = 1 liter
   - Final volume (V_{final}) = 10 liters
   - Constant external pressure (P_{ext}) = 1 bar
3. **Calculate Change in Volume (ΔV):** $\Delta V = V_{final} - V_{initial}$ = 10 liters − 1 liter = 9 liters
4. **Calculate Work Done in Liter Bar:** $W = -(1 \text{ bar}) \times (9 \text{ liters}) = -9$ liter bar
5. **Convert Work Done to Joules:** Using the conversion factor 1 liter bar = 100 joules: $W = -9 \text{ liter bar} \times \frac{100 \text{ joules}}{1 \text{ liter bar}} = -900$ joules

Conclusion

The correct work done is -900 joules. The negative sign indicates that work is done *by* the system (gas) on the surroundings during expansion. If only the magnitude is asked, the answer is 900 joules.

Work Done During Gas Expansion - Problem Solving

pressure of 10^{-5} Newton per meter square. What is the work done?

Formula for Work Done

The work done (W) during expansion against a constant external pressure is given by: $W = -P_{ext} \times \Delta V$ Where:

- P_{ext} is the constant external pressure.
- ΔV is the change in volume, calculated as $V_{final} - V_{initial}$.

Given Values

- Initial Volume ($V_{initial}$) = 10^{-3} m^3
- Final Volume (V_{final}) = 10^{-2} m^3
- Constant External Pressure (P_{ext}) = 10^{-5} N/m^2
- Temperature = 300 K (Note: Temperature is given but not directly used in this specific work done calculation for constant pressure expansion).

Step-by-Step Solution

1. **Calculate the change in volume (ΔV):** $\Delta V = V_{final} - V_{initial}$ = 10^{-2} m^3 – 10^{-3} m^3 To simplify, take 10^{-2} common: $\Delta V = 10^{-2}(1 - 10^{-1}) = 10^{-2}(1 - 0.1) = 10^{-2}(0.9) = 0.9 \times 10^{-2}$ m^3

2. **Substitute values into the work done formula:** $W = -P_{ext} \times \Delta V$ $W = -(10^{-5}$ $N/m^2) \times (0.9 \times 10^{-2}$ $m^3)$ $W = -0.9 \times 10^{-7}$ $N \cdot m$

3. **Convert units to Joules:** Since 1 Newton-meter (N·m) = 1 Joule (J): $W = -0.9 \times 10^{-7}$ J

Addressing the Discrepancy in the Question

The calculated answer (-0.9×10^{-7} J) does not match typical options for such problems, which often involve larger values like 900 J. This suggests a potential error in the given external pressure value in the problem statement.

If the external pressure was intended to be 10^5 N/m^2 instead of 10^{-5} N/m^2, the calculation would be:

1. **Recalculate with corrected pressure:** P_{ext} = 10^5 N/m^2 $W = -(10^5$ $N/m^2) \times (0.9 \times 10^{-2}$ $m^3)$ $W = -0.9 \times 10^{(5-2)}$ $N \cdot m$ $W = -0.9 \times 10^3$ $N \cdot m$ $W = -900$ $N \cdot m$

2. **Convert to Joules:** $W = -900$ J

Conclusion

Irreversible Work

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First Law of Thermodynamics

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First Law of Thermodynamics and Internal Energy for Different Processes

Introduction to the First Law of Thermodynamics

The first law of thermodynamics states that energy cannot be created or destroyed, only transferred or changed from one form to another. For a thermodynamic system, this is expressed as:

$$\Delta U = Q + W$$

Where:

- ΔU is the change in internal energy of the system.
- Q is the heat added to or removed from the system.
- W is the work done on or by the system.

We will now examine how this equation applies to different types of thermodynamic processes.

Isochoric Process

An isochoric process is a thermodynamic process in which the volume of the system remains constant.

- **Condition:** Change in volume (ΔV) is equal to 0.
- **Work Done:** Since work (W) is defined as $-P_{ext}\Delta V$, if $\Delta V = 0$, then the work done is also 0.
  - This holds true for both reversible and irreversible processes.
- **Internal Energy Change:** Substituting $W = 0$ into the first law of thermodynamics: $\Delta U = Q + 0$ $\Delta U = Q$ In an isochoric process, any change in internal energy is solely due to the heat absorbed or dissipated by the system.

system and its surroundings.

- **Condition:** Heat change (Q) is equal to 0.
- **Internal Energy Change:** Substituting $Q = 0$ into the first law of thermodynamics: $\Delta U = 0 + W$ $\Delta U = W$ In an adiabatic process, the change in internal energy is equal to the work done on or by the system.

Isothermal Process

An isothermal process is a thermodynamic process in which the temperature of the system remains constant.

- **Condition:** Change in temperature (ΔT) is equal to 0.
- **Internal Energy of Gases:** For gases, most of the internal energy is in the form of kinetic energy of the molecules.
  - Kinetic energy is a direct measure of temperature. As temperature increases, kinetic energy increases, and vice versa.
  - The relationship is often given as $KE = \frac{3}{2}RT$ for an ideal gas.
- **Internal Energy Change:** Since $\Delta T = 0$, the change in kinetic energy is also 0. As the internal energy of a gas is predominantly kinetic energy, a zero change in temperature implies a zero change in internal energy. $\Delta U = 0$
- **First Law Application:** Substituting $\Delta U = 0$ into the first law of thermodynamics: $0 = Q + W$ $Q = -W$ In an isothermal process, the heat absorbed by the system is equal to the negative of the work done (i.e., heat absorbed is equal to the work done by the system).

Isobaric Process

An isobaric process is a thermodynamic process in which the pressure of the system remains constant.

- **Condition:** Pressure (P) is constant, meaning $\Delta P = 0$.
- **Work Done:** Since pressure is constant, work done is typically irreversible work, calculated as $W = -P_{ext}\Delta V$.
- **Internal Energy Change:** The first law of thermodynamics remains in its general form for an isobaric process: $\Delta U = Q + W$ There is no simplification to the equation itself, as both heat and work can be non-zero.

Cyclic Process

A cyclic process is a thermodynamic process in which the system returns to its initial state after a series of changes.

- **Heat and Work:** Although $\Delta U = 0$, both heat (Q) and work (W) are path functions and are generally not zero in a cyclic process.
  - $Q \neq 0$
  - $W \neq 0$
- **First Law Application:** From $\Delta U = Q + W$ and $\Delta U = 0$: $0 = Q + W$ $Q = -W$ In a cyclic process, the net heat exchanged with the surroundings is equal to the negative of the net work done by the system.

Adiabatic Free Expansion

Understanding Adiabatic Free Expansion

Adiabatic free expansion is a specific thermodynamic process characterized by two key conditions:

1. **Adiabatic Process:** This means that there is no heat exchange between the system and its surroundings. Mathematically, this is represented as $Q = 0$.
2. **Free Expansion:** This refers to the expansion of a gas into a vacuum. In such a scenario, there is no external pressure acting on the gas. Therefore, the external pressure ($P_{external}$) is equal to zero.

Work Done During Free Expansion

The work done (W) during an expansion process is generally given by the formula:

$$W = -P_{external} \times \Delta V$$

Where:

- $P_{external}$ is the external pressure.
- ΔV is the change in volume.

Since free expansion occurs in a vacuum, $P_{external} = 0$. Consequently, the work done during free expansion is also zero:

$$W = -0 \times \Delta V = 0$$

Change in Internal Energy (ΔU)

According to the First Law of Thermodynamics, the change in internal energy (ΔU) of a system is given by:

$$\Delta U = Q + W$$

For an adiabatic free expansion, we have established that:

Substituting these values into the First Law of Thermodynamics equation:

$\Delta U = 0 + 0 \Delta U = 0$

Therefore, for an adiabatic free expansion of a gas, the change in internal energy is zero.

Comparison with Isobaric Process

It's important to distinguish this from an isobaric process, where the pressure remains constant ($\Delta P = 0$), but the pressure itself is not zero (e.g., one atmospheric pressure). In an isobaric process, work is generally not zero unless there is no change in volume.

Problem: Internal Energy Change During Spring Compression

Question

During compression of a spring, the work done is 10 kilojoules, and 2 kilojoules escaped to the surroundings as heat. The change in internal energy is:

Understanding the System and Work Done

- **System:** The spring is considered the system.
- **Work Done:** Work is done *on* the spring because it is being compressed. A spring does not compress on its own. When work is done *on* the system, the work value is positive.
  - Therefore, work done (W) = +10 kJ.

Understanding Heat Transfer

- **Heat Escaped:** 2 kilojoules of heat escaped to the surroundings, meaning heat moved *out* of the system.
- When heat is lost by the system, the heat value is negative.
  - Therefore, heat (Q) = −2 kJ.

Applying the First Law of Thermodynamics

- The change in internal energy (ΔU) is given by the First Law of Thermodynamics: $\Delta U = Q + W$
- Substitute the values of Q and W: ΔU = (−2 kJ) + (+10 kJ) ΔU = 8 kJ

Final Answer

The change in internal energy is 8 kJ.

Notes

A gas has constant pressure in the system. There is a loss of 45 joules of heat to the surrounding. 450 joules of work is done on the system. Find the change in internal energy during the process.

Solution

To find the change in internal energy (ΔU), we use the first law of thermodynamics, which states:

$$\Delta U = Q + W$$

Where:

- ΔU is the change in internal energy.
- Q is the heat exchanged with the surroundings.
- W is the work done.

Step-by-step Derivation:

1. **Identify the heat exchange (Q):**
    - The problem states there is a "loss of 45 joules of heat to the surrounding".
    - When heat is lost *by* the system, Q is negative.
    - Therefore, $Q = -45$ J.
2. **Identify the work done (W):**
    - The problem states "450 joules of work is done *on* the system".
    - When work is done *on* the system, W is positive.
    - Therefore, $W = +450$ J.
3. **Calculate the change in internal energy (ΔU):**
    - Substitute the values of Q and W into the first law of thermodynamics equation: $\Delta U = Q + W$ $\Delta U = (-45\text{ J}) + (450\text{ J})$ $\Delta U = 405$ J

Final Answer

The change in internal energy during the process is 405 J.

Problem: Change in Internal Energy Calculation

Question

Solution

To calculate the change in internal energy (ΔU), we use the first law of thermodynamics: $\Delta U = Q + W$ Where:

- Q is the heat exchanged.
- W is the work done.

Step 1: Calculate Heat (Q) The system is cooled by removing 124 joules of heat. When heat is removed from the system, Q is negative. $Q = -124$ J

Step 2: Calculate Work Done (W) Work done at constant pressure is given by: $W = -P_{\text{ext}}\Delta V$ First, convert the external pressure from torr to atmosphere: Given: 1 atm = 760 torr $P_{\text{ext}} = 1520$ torr $= \frac{1520}{760}$ atm = 2 atm Next, calculate the change in volume (ΔV). The gas contracts from 377 ml to 177 ml. $\Delta V = V_{\text{final}} - V_{\text{initial}} = 177$ ml $- 377$ ml $= -200$ ml Convert ΔV from milliliters to liters: $\Delta V = -200$ ml $= -0.2$ L Now, calculate the work done: $W = -(2 \text{ atm}) \times (-0.2 \text{ L}) = +0.4$ L atm

Step 3: Convert Work Done to Joules We know that 1 L atm ≈ 101.3 J. For approximation, we can use 1 L atm ≈ 100 J. $W = 0.4$ L atm $\times 100 \frac{\text{J}}{\text{L atm}} = 40$ J

Step 4: Calculate Change in Internal Energy (ΔU) Now, substitute the values of Q and W into the first law of thermodynamics equation: $\Delta U = Q + W = -124$ J $+ 40$ J $= -84$ J

Based on the options provided in the original problem, the closest answer is option B, which is −83.48 J. The slight difference might be due to using a more precise conversion factor for L atm to Joules (e.g., 101.3 J instead of 100 J). If we use 101.3 J: $W = 0.4$ L atm $\times 101.3 \frac{\text{J}}{\text{L atm}} = 40.52$ J $\Delta U = -124$ J $+ 40.52$ J $= -83.48$ J

Final Answer

The change in internal energy is −83.48 J.

Introduction and Welcome

Welcome and Student Introductions

The session began with a brief interaction where students introduced themselves by mentioning their home states. The instructor encouraged students to feel comfortable and even eat snacks during the class, emphasizing that hunger should not hinder learning. States mentioned by students included:

- Madhya Pradesh (MP)
- Odisha

- Karnataka
- Andhra Pradesh (AP)
- Telangana
- Bihar

Instructor's Background

The instructor shared personal details, clarifying that while she is from Chennai and was born and brought up in Tamil Nadu, her mother tongue is Telugu. She also acknowledged students who recognized her from Vedantu Master Tamil videos.

Learning Hindi

The instructor mentioned that she is actively learning Hindi to better communicate with students.

Class Approach

To prevent mental fatigue, the instructor stated her intention to intersperse casual conversations with the subject matter, avoiding a continuous, intense lecture format. This approach aims to provide breaks and maintain student engagement throughout the class.

Calculation of Work in Thermodynamics

Introduction to Work Calculation

Work calculation is a fundamental aspect of thermodynamics. We have previously discussed the calculation of irreversible work.

Work Done During Compression and Expansion

- **Compression Work:** When a system undergoes compression, work is done *on* the system. According to thermodynamic conventions, work done *on* the system is considered **positive**.
- **Expansion Work:** When a system undergoes expansion, work is done *by* the system. Work done *by* the system is considered **negative**.

Reversible vs. Irreversible Work

Introduction

We will now explore the calculation of work for both irreversible and reversible processes. Understanding the distinction between these two types of work is crucial in thermodynamics.

Reversible Work

- **Analogy: Bicycle Pump**
  - Imagine a bicycle pump. When you apply pressure, the piston moves inward, compressing the air.
  - However, if you release the pressure, the piston doesn't stay in the compressed position; it moves back outwards, though not necessarily to its exact original position.
  - This movement back and forth, where the system (air in the pump) responds to changes in external pressure, is an example of a reversible process.
  - In a reversible process, the change in volume (ΔV) and the external pressure are continuously changing. The work done at each instant is infinitesimally small, as the system is always close to equilibrium.
  - When drawing a PV graph for reversible work, it would be a continuous curve, reflecting the continuous change in pressure and volume.

Irreversible Work

Irreversible work, in contrast, is a process that cannot be reversed by small changes in conditions. Once an irreversible process occurs, the system cannot return to its initial state without some permanent change in the surroundings.

- **Analogy: Syringe**
  - Consider a syringe. When you push the plunger, it moves inward, expelling the liquid.
  - Once pushed, the plunger typically stays in that position; it does not spontaneously move back out.
  - If a syringe were to behave reversibly (like the bicycle pump), when a doctor injects medicine, the plunger would move back, potentially drawing blood back into the syringe, which is undesirable and doesn't happen in a typical injection.
  - Therefore, the action of a syringe is a good example of an irreversible process, where the system does not return to its original state on its own after the external force is removed.

Work Done in Irreversible Processes

Calculating Pressure-Volume Work for Irreversible Processes

In an irreversible process, the external pressure changes discontinuously. This means that at each step, a different external pressure is applied, leading to a corresponding change in volume. When plotting this on a P-V diagram, instead of a smooth curve, you would observe a series of steps, where each step represents a specific external pressure and the resulting volume change.

Graphical Representation and Work Calculation

To calculate the work done in an irreversible process, we consider the area under the P-V curve. Since the process involves discrete changes, the graph would appear as a series of connected

Notes

Integration for Work Calculation

For a continuous process or to approximate the work done by connecting these points with a curve, the work done (W) can be calculated by integrating the pressure with respect to volume. This is a fundamental concept in thermodynamics, often encountered in physical chemistry, especially in topics like chemical kinetics.

Mathematically, the work done is given by the integral of Y dx, where Y represents pressure and dx represents the change in volume. In the context of pressure-volume work, this translates to:

$$W = -\int_{V_{initial}}^{V_{final}} P_{external} dV$$

Here:

- W is the work done.
- $P_{external}$ is the external pressure.
- dV is the infinitesimal change in volume.
- $V_{initial}$ is the initial volume.
- V_{final} is the final volume.

The negative sign is a convention indicating that work done *by* the system is negative (expansion) and work done *on* the system is positive (compression).

Work Done in Isothermal Reversible Process

Derivation of Work Done Formula

For an isothermal reversible process, the work done can be calculated using the following steps:

1. **Starting Point:** We know that for an ideal gas, $PV = nRT$. Therefore, pressure P can be expressed as $P = \frac{nRT}{V}$.
2. **Work Formula:** The work done (W) is given by the integral of external pressure (P_{ext}) with respect to change in volume (dV): $W = -\int P_{ext} dV$ For a reversible process, $P_{ext} = P_{internal}$, so we can substitute $P = \frac{nRT}{V}$ into the work equation: $W = -\int_{V_i}^{V_f} \frac{nRT}{V} dV$ where V_i is the initial volume and V_f is the final volume.
3. **Constant Terms:** In an isothermal reversible process, temperature (T) is constant, the number of moles (n) is constant, and the gas constant (R) is constant. We can take these terms out of the integral: $W = -nRT \int_{V_i}^{V_f} \frac{1}{V} dV$
4. **Integration:** The integral of $\frac{1}{V} dV$ is $\ln(V)$: $W = -nRT[\ln(V)]_{V_i}^{V_f}$

$-nRT \ln\left(\frac{V_f}{V_i}\right)$ This is the formula for work done in an isothermal reversible process in terms of natural logarithm.

Conversion to Log Base 10

To convert from natural logarithm (ln) to logarithm base 10 (log), we multiply by 2.303:

$$W_{reversible} = -2.303nRT \log\left(\frac{V_f}{V_i}\right)$$

Work Done in Terms of Pressure

Since pressure and volume are inversely proportional ($P_1V_1 = P_2V_2 \implies \frac{V_f}{V_i} = \frac{P_i}{P_f}$), we can also express the work done in terms of initial pressure (P_i) and final pressure (P_f):

$$W_{reversible} = -2.303nRT \log\left(\frac{P_i}{P_f}\right)$$

These formulas are crucial for calculating work done in isothermal reversible processes.

Work Done in an Isochoric Process

Isochoric Process

An isochoric process is a thermodynamic process in which the volume of the system remains constant.

Work Done in Isochoric Process

For an isochoric process, the change in volume (ΔV) is equal to zero.

Since work done (W) is given by the formula $W = -P_{ext}\Delta V$, if $\Delta V = 0$, then the work done (W) in an isochoric process is also zero.

First Law of Thermodynamics for Isochoric Process

According to the first law of thermodynamics, $\Delta U = Q + W$. Since $W = 0$ for an isochoric process, the equation simplifies to: $\Delta U = Q$ This means that for an isochoric process, the change in internal energy of the system is equal to the heat exchanged with the surroundings.

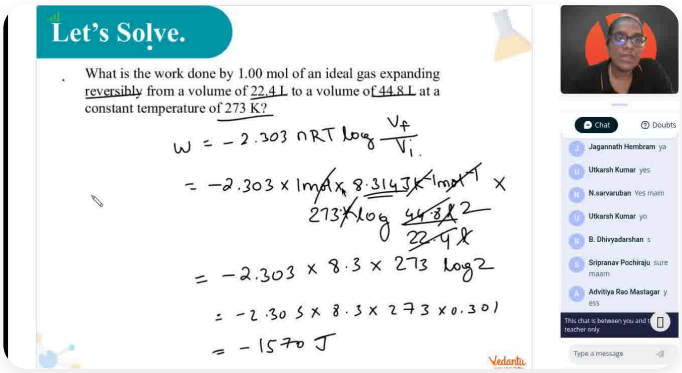
PV Graph for an Isochoric Process

In a Pressure-Volume (PV) graph, an isochoric process is represented by a vertical line. This is because the volume remains constant while the pressure can change.

- **Appearance:** A straight vertical line.

Notes

Problem: Work Done by Ideal Gas (Reversible Expansion)



Question

What is the work done by 1.00 mol of an ideal gas expanding reversibly from a volume of 22.4 L to a volume of 44.8 L at a constant temperature of 273 K?

Formula for Reversible Work Done

For a reversible expansion of an ideal gas, the work done (W) is given by the formula: $W = -2.303nRT \log \frac{V_f}{V_i}$ Where:

- n = number of moles
- R = gas constant
- T = temperature in Kelvin
- V_f = final volume
- V_i = initial volume

Given Values

- Number of moles (n) = 1 mol
- Initial volume (V_i) = 22.4 L
- Final volume (V_f) = 44.8 L
- Constant temperature (T) = 273 K
- Gas constant (R) = 8.314 J K^{-1} mol^{-1} (since the final answer is required in Joules)

Step-by-Step Solution

1. **Substitute the given values into the formula:** $W = -2.303 \times 1 \text{ mol} \times 8.314 \text{ J K}^{-1}\text{mol}^{-1} \times 273 \text{ K} \log \frac{44.8 \text{ L}}{22.4 \text{ L}}$
2. **Simplify the volume ratio:** The ratio $\frac{44.8 \text{ L}}{22.4 \text{ L}}$ simplifies to $\log 2$. $W = -2.303 \times 1 \text{ mol} \times 8.314 \text{ J K}^{-1}\text{mol}^{-1} \times 273 \text{ K} \log 2$

- 'L' in the numerator and denominator of the log term cancel out.

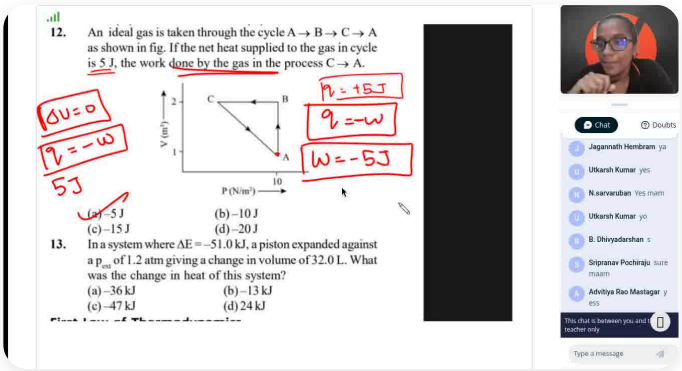
This leaves 'J' as the unit for work done.

4. **Use the approximate value for** $\log 2$: $\log 2 \approx 0.301$ $W = -2.303 \times 8.314 \times 273 \times 0.301$
5. **Calculate the final value:** $W \approx -2570$ J

Final Answer

The work done by the gas is approximately -2570 Joules. The negative sign indicates that work is done *by* the system (gas) on the surroundings during expansion.

Problem: Work Done in a Cyclic Process



Question

An ideal gas is taken through the cycle A→B→C→A as shown in the figure. If the net heat supplied to the gas in this cycle is 5 J, what is the work done by the gas in the process C→A?

(a) -5 J (b) -10 J (c) -15 J (d) -20 J

Solution

1. **Identify the Process:** The problem describes a cyclic process, A→B→C→A. This means the gas returns to its initial state after completing the cycle.

2. **First Law of Thermodynamics for Cyclic Process:** For any cyclic process, the change in internal energy (ΔU) is zero because the initial and final states are the same. According to the First Law of Thermodynamics: $\Delta U = Q + W$ Where:

   - ΔU is the change in internal energy.
   - Q is the heat supplied to the system.
   - W is the work done on the system.

3. **Given Information:**
    - Net heat supplied to the gas (Q) = +5 J (positive because heat is supplied *to* the gas).
4. **Calculate Work Done by the Gas:** Using the relationship $Q = W_{by}$ for a cyclic process: $W_{by} = Q$ W_{by} = +5 J The question asks for the work done *by* the gas in the process C→A. However, the provided solution in the slide and the verbal explanation directly relate the net heat supplied to the net work done *by* the gas over the *entire cycle*. The options provided are for the net work done. If the question specifically asks for work done in C→A, and the options are for the net work, there might be a slight ambiguity in the question phrasing versus the intended solution. Based on the provided solution steps and the options, it seems the question is implicitly asking for the *net work done by the gas over the entire cycle*.

    Therefore, the net work done by the gas over the cycle is: $W_{net,by} = Q_{net}$ = +5 J

    However, the provided solution in the slide and the verbal explanation state $W = -5J$. This implies that W here refers to the work done *on* the gas, or that the question is asking for the work done *on* the gas. Let's re-evaluate based on the common convention where W is work done *by* the gas, and the First Law is $Q = \Delta U + W_{by}$.

    If $Q = \Delta U + W_{by}$: For a cyclic process, $\Delta U = 0$. So, $Q = W_{by}$. Given Q = +5 J, then W_{by} = +5 J.

    The slide shows: $\Delta U = 0$ $q = -w$ (Here, q is heat supplied, w is work done *by* the gas) q = $+5J$ $W = -5J$

    This implies that the convention used in the slide is $Q = \Delta U - W_{by}$ or $Q = \Delta U + W_{on}$. If $Q = \Delta U + W_{on}$, then $Q = W_{on}$ for a cyclic process. So, $W_{on} = +5J$. If $W_{on} = +5J$, then $W_{by} = -W_{on} = -5J$.

    Therefore, the work done *by* the gas in the cycle is -5 J.

Final Answer

The work done by the gas in the cycle is -5 J.

The final answer is $\boxed{\text{-5 J}}$

Work Done in an Adiabatic Process with Constant External Pressure

Problem Statement

A real gas undergoes an adiabatic process from an initial state of 2 bar, 40 liters, and 300 Kelvin to a final state of 4 bar, 30 liters, and 300 Kelvin, against a constant external pressure of 4 bar. Calculate the work done for this process.

meaning $Q = 0$.

- **Work Done Against Constant External Pressure (Irreversible Process):** When a gas expands or compresses against a constant external pressure, the work done (W) is given by the formula: $W = -P_{ext} \times \Delta V$ Where:
  - P_{ext} is the constant external pressure.
  - ΔV is the change in volume, calculated as $V_{final} - V_{initial}$.

Step-by-Step Solution

1. **Identify Given Values:**
   - Initial Pressure ($P_{initial}$) = 2 bar
   - Initial Volume ($V_{initial}$) = 40 liters
   - Initial Temperature ($T_{initial}$) = 300 K
   - Final Pressure (P_{final}) = 4 bar
   - Final Volume (V_{final}) = 30 liters
   - Final Temperature (T_{final}) = 300 K
   - Constant External Pressure (P_{ext}) = 4 bar
2. **Calculate the Change in Volume (ΔV):** $\Delta V = V_{final} - V_{initial}$ = 30 liters − 40 liters = −10 liters
3. **Calculate the Work Done (W):** Using the formula $W = -P_{ext} \times \Delta V$: W = −(4 bar) × (−10 liters) W = +40 liter bar
4. **Convert Liter Bar to Joules:** It is known that 1 liter bar = 100 Joules. W = 40 liter bar × $\frac{100\text{ J}}{1\text{ liter bar}}$ = 4000 J

Final Answer

The work done for the process is **+4000 Joules**.

Explanation of Pressure Change and Compression

- The problem states that the internal pressure of the gas changes from 2 bar to 4 bar, while the external pressure is constant at 4 bar.
- Initially, the internal pressure (2 bar) is less than the external pressure (4 bar). This pressure difference causes the piston to move, compressing the gas.
- The compression continues until the internal pressure of the gas becomes equal to the external pressure. In this case, the compression stops when the internal pressure reaches 4 bar (equal to P_{ext}), at which point the volume is 30 liters.

- The fact that the initial and final temperatures are the same (300 K) despite it being an adiabatic process might seem contradictory. However, for a real gas, the internal energy change is not solely dependent on temperature, and the problem specifies the initial and final states, allowing for the calculation of work done based on the constant external pressure.
- The key takeaway is that for irreversible processes against a constant external pressure, the work done is calculated using the external pressure, not the changing internal pressure of the gas.

Session Conclusion and Feedback

Session Summary

This session concluded with the instructor seeking feedback from students regarding the clarity and effectiveness of the lecture. The instructor noted that some students rated the session with 4 out of 5, prompting a discussion to understand if this indicated any lack of understanding or vagueness in the content.

Key Takeaways

Did you find these notes helpful?

- The instructor values student ratings as a primary method for gauging comprehension.
- Direct interaction was used to clarify the meaning behind the ratings.

Upcoming Topics

The instructor confirmed that future sessions would cover topics such as pressure conversion and gas constants, as requested by students.

Homework and Telegram Group

- Homework assignments will be shared in the dedicated Telegram group.
- Students who have not yet joined the Telegram group were encouraged to do so using the provided link.